

QTrak SAML SSO Configuration Guide

Introduction

This guide provides configuration instructions for integrating a SAML 2.0 Identity Provider (IdP) with QTrak. It covers required attribute mappings, role configuration, Identity Provider setup, and the information required to complete the integration.

Prerequisites

- Administrator access to your Identity Provider
- A completed QTrak SAML implementation request
- QTrak Organization ID
- QTrak Service Provider (SP) metadata
- Identity Provider metadata
- Access to the QTrak implementation team

Supported Identity Providers

- Microsoft Entra ID (Azure Active Directory)
- Okta
- Other SAML 2.0 compatible Identity Providers

Required SAML Attribute Mapping

IdP Attribute	QTrak Attribute	Required	Notes
Email	Email	Yes	Primary identifier
Name	Name	Yes	User display name
organizationID	organizationID	Yes	Static value provided by QTrak
RoleID	roleID	No	Optional if using the default role (Supervisor)

Optional Attributes

IdP Attribute	QTrak Attribute	Purpose
FirstName	FirstName	User information
LastName	LastName / SurName	User information

Role Configuration

Role ID	Role
2	Supervisor (default)
3	Searcher
4	Scanner Operator

Note: If no RoleID is supplied, users are assigned the Supervisor role by default.

Note: For detailed information about each role and its permissions, refer to the [QTrak Complete User Guide](#).

Group-Based Role Assignment (Optional)

Organizations may assign roles through Identity Provider security groups instead of specifying a static RoleID.

Microsoft Entra ID Configuration (Previously Azure AD)

Step 1 - Create Security Groups

Create the following security groups in your Microsoft Entra ID:

1. **QTrak-Supervisors**
 - a. **Description:** QTrak Supervisor Role Access
 - b. **Group Type:** Security
 - c. **Members:** Users requiring Supervisor level access
2. **QTrak-Searchers**

- a. **Description:** QTrak Searcher Role Access
- b. **Group Type:** Security
- c. **Members:** Users requiring Searcher level access

3. QTrak-Operators

- a. **Description:** QTrak Scanner Operator Role Access
- b. **Group Type:** Security
- c. **Members:** Users requiring Scanner Operator level access

Step 2 - Configure Group Creation in Entra ID

1. Navigate to Identity -> Groups -> All Groups
2. Select "New group"
3. Configure the group using the following settings:
 - a. **Group type:** Security
 - b. **Group name:** Use the naming convention described above.
 - c. **Description:** Enter an appropriate description.
 - d. **Membership type:** Assigned (recommended for role-based access)
4. Select "Create."
5. Repeat these steps for each required QTrak security group.
6. Add users to the appropriate group based on the QTrak role they require.

Step 3 - Configure Group Claims in Entra ID

1. Navigate to Identity -> Applications -> Enterprise applications.
2. Select your QTrak enterprise application.
3. Select Single sign-on.
4. Under Attributes & Claims, select Add a group claim.
5. Configure the following settings:
 - a. **Which groups associated with the user should be returned in the claim:**
Security groups
 - b. **Source attribute:** Group ID
 - c. **Name:** groups
 - d. **Namespace:** Leave blank (or configure a custom namespace if required)
6. Save the configuration.

Step 4 - Configure Role Mapping

Configure your SAML assertion to include group membership information and map it to the RoleID attribute using the following logic:

XML

```
<saml:Attribute Name="RoleID">
  <saml:AttributeValue>
    <!-- If user is a member of QTrak-Supervisors -->
    2
  </saml:AttributeValue>
</saml:Attribute>
```

Microsoft Entra Group	RoleID
QTrak-Supervisors	2
QTrak-Searchers	3
QTrak-Operators	4

Step 5 - Configure Conditional Access (Optional)

Conditional Access policies provide an additional layer of security by controlling how users access the QTrak application based on your organization's requirements. This is not required for QTrak SSO to function and is an optional Microsoft Entra ID feature.

1. Navigate to **Protection** → **Conditional Access**.
2. Select **Create new policy**.
3. Configure the policy as needed:
 - **Users or workload identities:** Select the users or security groups that require access to QTrak.
 - **Target resources:** Select your **QTrak** enterprise application.
 - **Conditions:** Configure any required conditions (for example, device compliance, trusted locations, or sign-in risk).
 - **Grant:** Configure the appropriate access controls, such as requiring multifactor authentication (MFA) or requiring a compliant device.
4. Review the policy settings.
5. Enable the policy and save your changes.

QTrak Service Provider Information

Sample values are shown below. QTrak will provide the actual values applicable to your environment.

Setting	Value
Hosted Authentication URL	<code>https://qtrakssso.auth.us-east-1.amazonaws.com/oauth2/authorize?client_id=2kkpa8ilsldcokuu716i6l86us&response_type=token&scope=aws.cognito.signin.user.admin+email+openid+profile&redirect_uri=https%3A%2F%2Fdevv2.qtrak.net%2Fsaml_signin</code>
Entity ID	<code>urn:amazon:cognito:sp:us-east-1_pjNPR49dK</code>
ACS URL	<code>https://qtrakssso.auth.us-east-1.amazonaws.com/saml2/idpresponse</code>

Testing the Configuration

1. Export your Identity Provider metadata.
2. Send the metadata to QTrak.
3. Receive the configured SSO Initiation URL.
4. Sign in using a test account.
5. Confirm the correct user role is assigned.
6. Verify access to the QTrak application.
7. Confirm the user can successfully access the appropriate QTrak features based on their assigned role.

Troubleshooting

Groups not appearing in the SAML assertion

1. Verify group claims are configured correctly.
2. Check that users are members of the appropriate groups.
3. Ensure groups are security groups, not distribution groups.

Incorrect role assignment

1. Verify group ID mapping in your SAML configuration.
2. Check that the RoleID attribute is correctly mapped.
3. Ensure only one primary group is assigned per user.

Default role assigned unexpectedly

1. Verify group membership in Microsoft Entra ID.
2. Check that group claims are being sent in SAML response.
3. Ensure proper group-to-role mapping logic is implemented.

Additional Resources

AWS Cognito - SAML Identity Providers
Microsoft Entra ID SAML documentation
Okta SAML documentation

Next Steps

Static Role Assignment

1. Configure required attributes.
2. Configure `RoleID`.
3. Export IdP metadata.
4. Send metadata to QTrak.
5. Test SSO.

Group-Based Role Assignment

1. Create security groups.
2. Configure group claims.
3. Configure group-to-role mapping.
4. Export IdP metadata.
5. Send metadata to QTrak.
6. Test SSO.